Chapter 10 Definite Integrals: Examples Worksheet

Questions

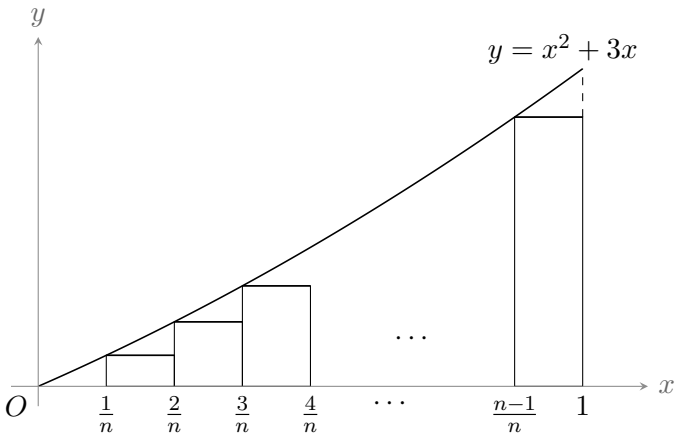
JC and A-level examination questions (H2 Mathematics 9758): three on the limit of a sum for Section 10.1, then one for each later section (Promotional Examination where available, otherwise Preliminary Examination). Questions are reproduced as set, with the original marks.

Q	Section	Source
1	10.1 The definite integral as a limit of a sum	DHS/2019/P1/Q10†
2	10.1 The definite integral as a limit of a sum	ACJC/2019/P1/Q6†
3	10.1 The definite integral as a limit of a sum	9758/2015/P1/Q3†
4	10.2 Area between a curve and the x-axis	NJC 2024 Promo Q3
5	10.3 Area between a curve and the y-axis	SAJC 2025 Prelim P1 Q7
6	10.4 Area between two curves	YIJC 2025 Prelim P1 Q12
7	10.5 Volume of revolution	NJC 2024 Promo Q2
8	10.6 Parametric curves: area	VJC 2025 Promo Q11(c), adapted

†From the Section 1 exercises in the teacher's own Definite Integrals notes.

Question 1 10.1

[DHS/2019/P1/Q10]



The diagram above shows part of the graph of $y = x^2 + 3x$, with rectangles approximating the area under the curve from $x = 0$ to $x = 1$. The area under the curve may be approximated by the total area, A, of $(n-1)$ rectangles each of width $\frac{1}{n}$. Given that

$$\sum_{r=1}^{n} r^2 = \frac{1}{6}n(n+1)(2n+1),$$

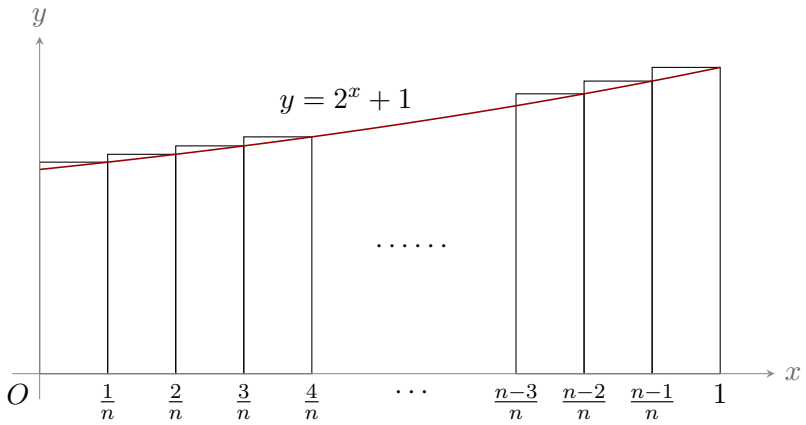
show that $A = \dfrac{(n-1)(11n-1)}{6n^2}$.

Explain briefly how the value of $\displaystyle\int_0^1 x^2 + 3x \ dx$ can be deduced from this expression, and hence find this value exactly without integration. **[6]**

Question 2 10.1

[ACJC/2019/P1/Q6]

The diagram below shows the graph of $y = 2^x + 1$ for $0 \leq x \leq 1$. Rectangles, each of width $\frac{1}{n}$, are also drawn on the graph as shown.



Show that the total area of all n rectangles, S_n, is given by

$$S_n = \frac{2^{1/n}}{n\left(2^{1/n} - 1\right)} + 1.$$

[3]

Find the exact value of $\lim_{n\to\infty} S_n$. **[2]**

Question 3 10.1

[9758/2015/P1/Q3]

(i) Given that f is a continuous function, explain, with the aid of a sketch, why the value of

$$\lim_{n\to\infty} \frac{1}{n}\left\{\mathrm{f}\left(\frac{1}{n}\right) + \mathrm{f}\left(\frac{2}{n}\right) + \cdots + \mathrm{f}\left(\frac{n}{n}\right)\right\}$$

is $\displaystyle\int_0^1 \mathrm{f}(x)\, dx$. **[2]**

(ii) Hence evaluate $\displaystyle\lim_{n\to\infty} \frac{1}{n}\left(\frac{\sqrt[3]{1} + \sqrt[3]{2} + \cdots + \sqrt[3]{n}}{\sqrt[3]{n}}\right)$. **[3]**

Question 4 10.2

[NJC 2024 Promo Q3]

(i) Given that k is a positive constant, sketch the curve E with equation $y = \left|x^2(x-k)\right|$, labelling the coordinates of the points where the curve cuts the axes. **[2]**

(ii) Hence, find, in terms of k, the area of the region(s) bounded by E and the x-axis for $0 \leq x \leq 2k$. **[3]**

Question 5 10.3

[SAJC 2025 Prelim P1 Q7]

(a) (i) On a single diagram, sketch the graphs of $y = \dfrac{3x}{x-1}$ and $y = 3x$, stating the equation(s) of any asymptote(s) and the coordinates of points of intersection. **[4]**

(ii) Hence find the area bounded by the curve $y = \dfrac{3x}{x-1}$ and the lines $y = 3x$ and $y = 8$. **[2]**

(b) A curve has the equation $y = \sqrt{x^2+1}$ where x is non-negative.

(i) With the aid of a sketch of the above curve, explain why the value of

$$S = \lim_{n\to\infty} \frac{1}{n}\left[1 + \sqrt{\frac{1}{n^2}+1} + \sqrt{\frac{4}{n^2}+1} + \ldots + \sqrt{\frac{(n-1)^2}{n^2}+1}\right] \text{ is } \int_0^1 \sqrt{x^2+1}\,dx.$$

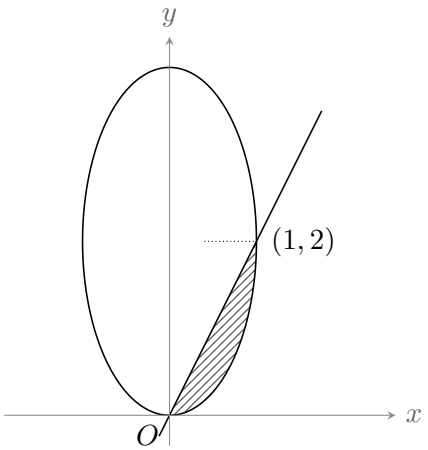
[4]

(ii) Hence find the value of S. **[1]**

Question 6 10.4

[YIJC 2025 Prelim P1 Q12]

(a) Use the substitution $x = \sin\theta$ to show that $\displaystyle\int_0^1 \sqrt{1-x^2}\,dx = \frac{\pi}{4}$. **[4]**



The diagram shows the region R bounded by the ellipse with equation $4x^2 + (y-2)^2 = 4$ and the line $y = 2x$. The ellipse and the line intersect at the origin and the point $(1, 2)$.

(b) Find the area of R, giving your answer in terms of π. **[4]**

(c) Find the volume of solid generated when R is rotated 2π radians about the x-axis. **[2]**

Question 7 **10.5** [NJC 2024 Promo Q2]

The region bounded by $y = x\sqrt{\sin x}$, the x-axis, and the lines $x = 0$ and $x = \pi$ is rotated through 2π radians about the x-axis. Find the exact volume of the solid generated. **[5]**

Question 8 **10.6** [VJC 2025 Promo Q11(c), adapted]

The parametric equations $x = t^2$, $y = 20te^{-t}$, for $0 \leq t \leq 6$, model a section of a roller coaster track; x is the horizontal distance and y the vertical distance from the ground, in metres. A wind-blocking panel is attached beneath the track. Its cross-sectional area is modelled by the area under the curve from $t = 0$ to $t = 6$. The area of the region under a parametric curve from $t = \alpha$ to $t = \beta$ is given by the formula $\displaystyle\int_{\alpha}^{\beta} y\left(\frac{dx}{dt}\right) dt$.

(i) Show that the cross-sectional area of the panel is $\displaystyle\int_{a}^{b} kt^2e^{-t}\, dt$, where a, b and k are integers to be determined. **[2]**

(ii) Find the exact area of the panel. **[4]**